

Section A: Conceptual Understanding

1. What is Ethical AI, and why is it essential in modern technological systems?

Ethical AI is the design and use of AI in a way that is fair, safe, transparent, and respects human rights. It is important because AI affects decisions about people's lives, jobs, healthcare, and privacy.

2. List and explain any four major ethical principles that guide AI development.

- **Fairness:** AI should not discriminate against people.
- **Transparency:** Users should know when AI is being used.
- **Accountability:** Developers and organizations are responsible for AI decisions.
- **Privacy:** Personal data should be protected and used responsibly.

3. What is the main objective of the IEEE Ethically Aligned Design framework?

Its goal is to help developers create AI systems that prioritize human well-being, ethics, and human rights.

4. Define the term algorithmic bias and provide one real-world example.

Algorithmic bias happens when an AI system produces unfair results because of biased data or design. Example: A hiring AI favors male applicants over equally qualified female applicants.

5. Explain the difference between transparency and explainability in AI.

Transparency means being open about how AI is developed and used. Explainability means being able to understand why the AI made a specific decision.

6. What role does accountability play in the ethical use of AI systems?

Accountability ensures that people or organizations are responsible for AI decisions and any harm the system causes.

7. Identify three international organizations involved in developing AI ethics guidelines.

- UNESCO
- OECD
- IEEE

8. What does UNESCO's Recommendation on the Ethics of Artificial Intelligence emphasize?

It emphasizes human rights, fairness, diversity, privacy, sustainability, and responsible AI development.

9. What are OECD's five key principles for trustworthy AI?

1. Inclusive growth and well-being
2. Human-centered values and fairness

3. Transparency and explainability
4. Robustness, security, and safety
5. Accountability

10. How does the EU AI Act classify and regulate AI systems based on risk?

It classifies AI into unacceptable, high, limited, and minimal risk categories. Higher-risk systems must meet stricter rules and oversight.

Section B: Applied and Scenario-Based

11. An AI recruitment tool consistently prefers male candidates. Which ethical principles are violated?

Fairness, non-discrimination, accountability, and transparency.

12. A self-driving car must choose between hitting one pedestrian or five. Which ethical dilemma does this represent?

The trolley problem.

13. In predictive policing, how can bias be introduced into the algorithm's training data?

If historical crime data already reflects biased policing, the AI learns and repeats those biases.

14. How might a "human-in-the-loop" system reduce ethical risks in automated decision-making?

A human reviews important AI decisions before they are finalized, helping catch mistakes or unfair outcomes.

15. Describe how data privacy laws such as GDPR affect the design of AI systems.

AI systems must protect personal data, collect only necessary information, obtain consent, and allow users to control their data.

16. A generative AI creates a fake video of a politician. What ethical issues arise here?

Misinformation, manipulation, privacy violations, reputational harm, and loss of public trust.

17. What are the ethical implications of AI-powered surveillance systems in public spaces?

They can reduce privacy, enable mass surveillance, and increase the risk of discrimination or misuse.

18. How can organizations ensure transparency in AI-driven financial decisions?

They should explain how AI makes decisions, document the process, and allow users to question or appeal decisions.

19. Discuss the ethical concerns surrounding AI in healthcare diagnostics.

Concerns include biased diagnoses, patient privacy, incorrect predictions, and determining responsibility for medical errors.

20. What measures can be implemented to make AI datasets more representative and fair?

Use diverse datasets, remove biased data, regularly test for bias, and continuously update the data.

Section C: Reflective and Future-Oriented

21. Do you think AI can ever be completely unbiased? Justify your answer.

No. Since AI learns from human-created data, some bias will always exist. The goal is to reduce bias as much as possible.

22. How can ethical frameworks ensure responsible innovation in AI?

They provide guidelines that encourage fairness, transparency, accountability, privacy, and human oversight during development.

23. Should governments enforce stricter AI regulations, or should industries self-regulate? Discuss.

A combination of both works best. Governments create laws, while companies follow ethical standards and improve best practices.

24. What ethical challenges might Quantum AI pose in the future?

It could break current encryption, increase privacy risks, and make AI decisions even harder to understand.

25. How might Generative AI reshape intellectual property rights?

It raises questions about who owns AI-generated content and whether copyrighted material was used without permission.

26. What are the moral responsibilities of AI developers toward end-users?

They should build AI that is safe, fair, transparent, secure, and respects users' rights and privacy.

27. How can AI sustainability contribute to global ethical goals?

By reducing energy use, lowering environmental impact, and supporting sustainable development.

28. What future jobs might emerge to address ethical issues in AI systems?

AI Ethics Officer, AI Auditor, Algorithm Bias Analyst, AI Compliance Specialist, and AI Governance Manager.

29. In your opinion, what will be the most pressing AI ethics issue in 2035, and why?

The spread of AI-generated misinformation will likely be the biggest issue because it can influence elections, public opinion, and trust in information.

30. Reflect on your role as a future professional — how would you ensure ethical practices in AI development?

I would follow ethical guidelines, test AI for bias, protect user privacy, document how the AI works, and make sure humans can review important decisions.